

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(a) | Bacteria <b>and</b> Archaea ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 |
| 4(b) | <p><i>any <b>four</b> from:</i></p> <ol style="list-style-type: none"> <li>1 obtain (specific restriction endonuclease) gene, qualified ;<br/>e.g. gene extracted from the, DNA / genome, of E. coli / bacteria<br/>gene synthesised from the mRNA of E. coli / bacteria<br/>gene synthesised from nucleotides</li> <li>2 PCR to make more copies of, gene / DNA ;</li> <li>3 cut a plasmid, with a restriction enzyme / to give sticky ends ;</li> <li>4 combine / mix / join, gene with plasmid using ligase ;</li> <li>5 add plasmid to, E. coli / bacteria ; <b>A</b> transformation of E. coli</li> <li>6 use marker genes to, select / identify, bacteria / E. coli, with recombinant plasmids ;</li> <li>7 <i>ref.</i> culture bacteria / use fermenter ;</li> <li>8 <i>ref.</i> extraction / purification (of enzyme) ;</li> <li>9 AVP ; e.g.heat shock / calcium ions / electroporation (detail mp5)<br/>complementary sticky ends anneal / make phosphodiester bonds (detail mp4)</li> </ol> | 4 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(c)     | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 (data stored is) a large amount of, amino acid / nucleotide / base / DNA, sequences ;</li> <li>2 share data / access data quickly ;</li> <li>3 compare (amino acid / nucleotide / base / DNA) sequences ;</li> <li>4 use to, predict / visualise / model, the structure of the enzyme ;</li> <li>5 identify restriction endonuclease to use<br/> <b>or</b><br/> find / discover, new restriction endonucleases ;</li> </ol> | <b>3</b> |

| Question | Answer                |                       |                                                                                 |                                                                    | Marks |   |
|----------|-----------------------|-----------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------|-------|---|
| 4(d)     | any <b>four</b> from: |                       |                                                                                 |                                                                    |       | 4 |
|          |                       | feature               | electrophoresis                                                                 | chromatography                                                     |       |   |
|          |                       | 1 separation role     | both – are separating techniques<br>or<br>both – involve movement of substances |                                                                    | ;     |   |
|          |                       | 2 molecule separated  | DNA / RNA / proteins                                                            | pigments / named pigment / amino acids                             | ;     |   |
|          |                       | 3 medium              | gel / named gel                                                                 | paper / glass / silica gel / named medium                          | ;     |   |
|          |                       | 4 separated by        | (electric) current / electric field / electricity                               | solvent / named solvent                                            | ;     |   |
|          |                       | 5 separated due to    | charges / polarity / lengths / mass                                             | solubilities / mass                                                | ;     |   |
|          |                       |                       | both – separate by mass                                                         |                                                                    |       |   |
|          |                       | 6 visualisation using | dyes / ethidium bromide / UV light                                              | different colours of pigments / dyes not added / UV light not used | ;     |   |
|          |                       |                       | both - dyes can be added (if TLC)                                               |                                                                    |       |   |
|          |                       | 7 identification by   | use, DNA ladder / known length of DNA<br>or<br>use of DNA probe                 | use R <sub>f</sub> value                                           | ;     |   |
|          |                       |                       | both - use of, references / known values                                        |                                                                    |       |   |